

Adyanth S

Machine Learning Engineer

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Portfolio | LinkedIn | GitHub | Kaggle

SUMMARY

Final-year B.Tech AI/ML student. I build clinical software end to end: **FHIR R4** hospital systems, **hybrid RAG** over locally hosted LLMs, and **QLoRA** fine-tuning in PyTorch. Looking for AI/ML engineering internships and entry-level roles.

EDUCATION

PES University, Bengaluru

2023 – 2027 (Expected)

B.Tech, Artificial Intelligence and Machine Learning

EXPERIENCE

PES University, Dept. of Artificial Intelligence & Machine Learning

Aug 2026 – Present

Teaching Assistant, Prompt Engineering & Optimization for Generative AI (UE24AM343AB5)

Bengaluru, India

- Selected by Dr. Arti Arya (Chairperson, AI & ML) to support a course on prompt engineering, optimization and **LLM evaluation**.

TECHNICAL SKILLS

Languages: Python, C, SQL, JavaScript

ML & AI: PyTorch, Hugging Face, TensorFlow, Scikit-learn, LoRA / QLoRA fine-tuning, RAG, hybrid retrieval (dense + BM25, RRF), multi-agent systems, NLP, RLHF, prompt engineering

Backend & Infra: FastAPI, Flask, LangChain, Ollama, PostgreSQL, Elasticsearch, Milvus, ChromaDB, MinIO, Docker, Nginx, Git, CUDA, Linux, Vercel

Healthcare Standards: FHIR R4, ICD-10, clinical audit trails, role-based access control

Quantum ML: Qiskit, quantum kernels, QSVM, hybrid quantum-classical models

PROJECTS

Clinical Platform: Hospital Information System and ED Triage Console

Live Demo | HIS Repo | Triage Repo

FastAPI, Flask, PostgreSQL, Elasticsearch, Milvus, MinIO, SQLite / Turso (libSQL), Llama 3.1 via Ollama, Docker, Nginx, FHIR R4, Vercel

- Built a hospital information system covering EHR, clinical workflows and ML decision support on a polyglot backend (**PostgreSQL**, **Elasticsearch**, **Milvus**, **MinIO**) behind a **FastAPI** gateway, with **FHIR R4** interoperability.
- Deployed the Emergency Department console on the same stack: **5-level triage**, treat-first registration, the Indian **medico-legal (MLC)** workflow with police intimation, and **six disposition pathways**, covered by **106 automated checks**.
- Wrote a **hybrid RAG** engine (dense + BM25, reciprocal-rank fusion, reranking, citation validation) over a locally hosted **Llama 3.1** so clinical answers cite their source instead of hallucinating.
- Locked down access with a tamper-evident **append-only SHA-256 audit chain** (database triggers blocking UPDATE and DELETE) and server-side RBAC across **5 clinical roles**.

Multi-Agent Medical Severity Classification

BioMistral-7B, MedAlpaca-13B, QLoRA, PyTorch, Hugging Face

- Built a multi-agent LLM pipeline (**BioMistral-7B** and **MedAlpaca-13B**) that sorts patient symptoms into **4 severity tiers**, with a confidence-aware hard-flag override for red-flag cases, reaching **92% accuracy**.
- Fine-tuned both models with **4-bit QLoRA** (rank 16, alpha 32) to fit consumer GPU memory.

NexusGuard: AI Governance and Access Control Platform

Hackathon

Llama 3.1 8B, ChromaDB, FastAPI, Ollama, RAG pipeline, Spring Boot, React

- Built **Need-Role-Level (NRL) access control** for LLM queries: every question is classified against the user's role before retrieval, so the model never sees documents the user cannot read. Owned the **FastAPI** service and the classifier over a **150-document ChromaDB** RAG pipeline.
- Fine-tuned Llama 3.1 8B for the classification step, found it **overfitting** the small role set, and replaced it with a structured system-prompt classifier that generalized better.

Additional Projects: **Quantum Kernel ECG Classification:** QSVM with a physiology-aware feature map and multi-lead entanglement (Qiskit), beating classical kernel baselines on noisy, low-sample signals. **Neural Network Weight Denoising:** restored corrupted parameters in an LSTM encoder-decoder translator (PyTorch). **Age Prediction from Gene Expression:** ElasticNet regression on GSE113957.

ACHIEVEMENTS

- 4th of 60 teams**, Praxis Hackathon.
- Runner-up, IDEATHON.** Built a quadriplegia assistance system for mobility and independence.
- 2nd place**, district-level mathematics competition, SDM College.